

How the equal-characteristic Lech proof candidate emerged

A reconstruction of elapsed time, the main mathematical breakthroughs, and the role played by the user prompt.

Status qualification. In this report, "proof" means the first complete candidate that survived several internal independent audits. It is not external peer validation or community acceptance. The argument covers equal characteristic; mixed characteristic remains unresolved.

37 h 54 m

FROM THE STRUCTURAL PROMPT

Sep 3, 02:31 to Sep 4, 16:25 PDT

54 h 23 m

FROM THE FIRST LECH SESSION

Sep 2, 10:03 to Sep 4, 16:25 PDT

The task clock had already exceeded 33 active hours two hours before the candidate appeared, so roughly 35 active elapsed hours went into the structural continuation. Exact cumulative agent-hours are unavailable because specialized agents worked in parallel and one service interruption was recorded.

The final 70 minutes were a synthesis burst after nearly two days of incubation. They should not be read as a 70-minute proof.

The logical spine

1

Dutta target

Turn a flat map into one numerical inequality for a short complex.

2

Graded depth

Use categorical pairings to convert deep columns into a sharp length bound.

3

Sparse stability

Charge only the shallow columns, independently of ranks and shifts.

4

Local transfer

Globalize jets and use periodic boxes to retain exact mass.

5

Singular bridge

Compress through a strict rank-e(D) lattice and Frobenius.

6

Lech

Pass from the Dutta bound to characteristic p, then spread to characteristic zero.

Timeline of ideas and breakthroughs

Sep 2 10:09 PDT	The exact Dutta target was identified. The Iyengar-Ma-Walker reduction turns a flat map into a short complex F over a possibly singular ring T, with $\chi_{\infty}(F) = e(S)$ and $e(T) = e(R)$. The north-star inequality became $\chi_{\infty}(F) \geq e(T)$, rather than a direct comparison of flat maps.
Sep 3 02:31 PDT	The investigation was deliberately reset at the structural level. The new prompt required at least six active hours, persistence until a proof or counterexample, and a larger-picture approach instead of more finite case analysis. The first new note, five minutes later, explicitly records that pivot.
Sep 3 06:50 PDT +4 h 19 m	First core engine: the graded large-gap theorem. For an actual finite-length graded resolution whose nonzero differential entries are N-deep, the Eisenbud-Erman categorical pairing preserves the depth condition. Supernatural bundles support the boundary facets of the Boij-Soderberg cone of pure diagrams with gaps at least N. This yields $\text{length}(M) \geq N^d \beta_i(M) / \text{binom}(d,i)$.
Sep 3 late morning	False numerical routes exposed the missing information. Formal Betti tables could satisfy separate coordinate, moment, and Tor tests while violating the desired bound. Exactness and compatibility between consecutive differential matrices ruled them out. The lesson was decisive: a proof had to retain actual column-level and higher-map information.
Sep 3 15:35 PDT +13 h 04 m	A strict multiplicity lattice bridged singular rings to regular ones. Inside a complete characteristic-p ring D, a strict free lattice of rank exactly $e(D)$ was constructed over a regular parameter subring A. Frobenius makes the compressed matrices $q - O(1)$ deep over A, while inclusion and retraction have a fixed scalar composite. This preserved the multiplicity coefficient that the final inequality needed.
Sep 3 night - Sep 4, 15:15	The graded-to-local and derived-to-actual bottlenecks dominated the next day. Passing to an associated graded object created shallow cancellations; scalar-null comparisons produced derived complexes rather than actual module resolutions; small boundary rank did not imply a small degree moment. Multiple Ulrich, absolute-Tor, deformation, and counterexample routes were rejected. At 15:15 the contemporaneous checkpoint still said: no proof.

The decisive synthesis window

Between 15:38 and 16:25 PDT on September 4, several previously separate components locked together. The work overlapped across agents: this was collaborative synthesis, not one sequential derivation.

- 15:38-15:56

A robust one-variable gap functional controlled exceptional blocks.
Homogeneous Smith form showed that every potentially negative barcode is detected by a restricted differential rank. A scalar-null comparison cone costs only boundary-size residue Betti mass after coordinate restriction, while the principal block grows like t^d .
- 16:00

Derived comparison was converted into an actual minimal head resolution.
Rotating the small comparison cone produced a resolution containing the large deep block literally. A labeled unit-cancellation argument preserved all but a lower-order number of its deep columns and confined arbitrary behavior to a sparse exceptional set.
- 16:05

Periodic degree boxes solved the finite-support transfer problem.
The substitution $x_i \rightarrow x_i u_i$ produced L^d copies. Periodic boxes preserved exact length, open boxes produced a genuine grading, and a finite-length repair required only $O(L^{(d-1)})$ free terms. Crucially, the leading L^d boundary Euler moment was kept inside the exact inequality rather than discarded.
- 16:11

The decisive sparse-column inequality was proved.
Translated supernatural bundles isolate one adjacent homological row. A good N -deep column vanishes in the translated window; each bad column appears in only N translations. Peskine-Szpiro moment identities and an explicit nonnegative interpolation extract each β_r , giving a uniform error $C_d N^d B$ independent of ranks and shifts.
- 16:12-16:22

Prescribed-jet globalization transferred the result to local rings.
Generators were perturbed inside actual syzygy modules using a Forster-Swan incidence argument. Quillen-Suslin handled the last free syzygy, determinant normalization preserved its jet, and finite-field descent retained all counts. Taking the box limit for each fixed module gave the exact local sparse inequality.
- 16:21-16:25

Frobenius, Dutta multiplicity, and the flat-map reduction completed the chain.
Coordinate-power restriction balanced rank t^d against depth q/t and made scalar-cone errors lower order. The strict lattice supplied the factor $e(D)$; the Bhatt-Hochster-Ma estimate made positive Frobenius homology negligible. The all-row Dutta inequality followed, then IMW gave characteristic p and Ma's spreading argument gave characteristic zero.
- 16:25-17:07

The candidate and its first independent audits were assembled.
The complete proof candidate was created at 16:25:41. Two whole-chain audits existed by 16:30, followed by a third central-lemma audit and an adversarial transfer-interface review. No fatal gap was found.
- Sep 7

The argument matured into a manuscript and further end-to-end audits.
A clean manuscript was finalized, and fresh delegated and full adversarial rereads checked the hypotheses, normalizations, and published inputs again. These remain internal checks, not external certification.

What the prompt contributed

PROCESS-CAUSAL	NOT CONTENT-CAUSAL	COUNTERFACTUAL
The structural instruction changed the search regime. The first new note explicitly says it was following the request to stop extending finite case analyses. The persistence requirement kept the work running far beyond the first partial theorem and the six-hour minimum.	The prompt supplied none of the decisive mathematics: no categorical pairing, translated gap test, periodic box, prescribed jet, coordinate-power restriction, strict lattice, or Dutta composition. Those ingredients emerged from literature synthesis and failed-route pressure.	It is impossible to know whether the same proof would eventually have appeared anyway. A routine short response likely would have stopped with partial results. Another equally sustained structural investigation might have found a similar chain, but its timing and exact form were not inevitable.

Other process factors

Parallel specialization. Separate agents developed the scalar-cone and degree-box components, while root assembled the translated sparse inequality, coordinate-power step, and final reduction. Independent agents repeatedly reconstructed vulnerable interfaces rather than merely agreeing with a summary.

Resource constraints. The later instruction to avoid expensive scans reinforced the structural focus, but it did not technically enable the proof. The decisive window used symbolic arguments and primary-source reading; no CPU-intensive enumeration worker was running.

Verification discipline. The instruction not to overclaim an open problem materially improved the result's auditability. It led to same-day whole-chain checks and fresh end-to-end rereads on September 7. That strengthens confidence without substituting for external review.

Bottom line. The prompt did not hide a proof idea. It created the conditions in which one became substantially more likely: a long horizon, a structural objective, permission to reject many attractive dead ends, and a requirement to keep checking rather than declare victory early.

Evidence base

The workspace has no Git repository, so this chronology uses NTFS creation and modification times together with contemporaneous checkpoint files. The first candidate was later edited; the exact original text diff is therefore not recoverable. The following records support the timing and dependency chain:

- Research clock** - C:\Users\agerko\Documents\Lech\RESEARCH.md
Start times, duration milestones, and the shift between the two research sessions.
- Structural audit** - C:\Users\agerko\Documents\Lech\structural_root_audit_20260903.md
Contemporaneous accepted results, rejected shortcuts, active-time notes, and final dependency genealogy.
- First assembled candidate** - C:\Users\agerko\Documents\Lech\candidate_equal_characteristic_lech_proof_20260904.md
The complete six-stage proof chain created September 4 at 16:25 PDT.
- Full adversarial audit** - C:\Users\agerko\Documents\Lech\candidate_equal_characteristic_lech_full_adversarial_theorem_audit_20260907.md
Fresh reconstruction of the main arguments and published hypotheses.
- Manuscript** - C:\Users\agerko\Documents\Lech\output\paper\lech_equal_characteristic.tex
Polished equal-characteristic manuscript and its logical-spine summary.